WEEK 08 | CNN CLASSIFICATION II

Intern: Mehrana Rezaei | Group 1: Basic ML Knowledge | Main Quest: Convolutional Neural Networks

1. Objective

The practical goal was to extend Week 07's five-object YCB-Video (YCB-V) classification system from a compact convolutional neural network (CNN) to a stronger, transfer-learning-based alternative, then determine whether increased model capacity improves both object recognition and robustness. I retrained the original CompactCNN and fine-tuned an ImageNet-pretrained ResNet18 on the same cropped objects. I evaluated them on clean scenes and under changes in viewpoint, scale, occlusion and clutter, rather than relying on a single accuracy figure.

2. Relevant Main Quest Concepts

CNN course concept	Where it was applied in Week 08
Convolution, filters, feature maps and pooling	CompactCNN used three convolutional layers (32/64/96 channels), ReLU and pooling to learn spatial features from RGB crops.
Deep CNNs and residual networks	ResNet18 supplied a deeper backbone with residual connections, contrasted with the small custom CNN.
Transfer learning / fine-tuning	Loaded ImageNet pretrained weights; trained the new 5-class classification head, then unfroze layer4 and the head.
Regularization and generalization	Used dropout (0.30) and weight decay; compared train/validation curves and results on distinct scenes.
Optimization and model selection	Applied mini-batch cross-entropy training, Adam / AdamW, fixed seeds, validation-loss checkpointing and early stopping.
Model error analysis	Confusion matrices, per-class recall, controlled input stress tests and real BOP annotation strata located failure conditions.

3. Method

Dataset, split and preprocessing

The dataset was the YCB-V BOP'19 source subset (Hugging Face: bop-benchmark/ycbv), treated as an internal cropped-object classification benchmark. Five object IDs were retained: 4 tomato_soup_can, 3 sugar_box, 1 master_chef_can, 6 tuna_fish_can, and 12 bleach_cleanser. Candidate instances required a bounding-box side of at least 24 pixels and visibility fraction ≥0.35. Each ground-truth bounding box received 8% padding; the RGB crop was resized with aspect-ratio-preserving letterbox placement on a 64×64 gray canvas (fill 127).

Split	Scene IDs	Samples
Training	48, 49, 50, 53, 54, 56, 57, 58	877
Validation	52, 55	375
Original held-out test	51, 59	450

The scene-level split, five-class task and seed 42 were fixed before comparison. The test archive name is BOP19, but this was not an official BOP leaderboard evaluation; scenes 51 and 59 had already been seen in prior Week 07 analysis.

Models and important parameters

Parameter / stage	Retrained CompactCNN	Pretrained ResNet18
Architecture	Conv channels 32/64/96; adaptive 4×4 pool; dense 128; dropout 0.30	ResNet18 ImageNet DEFAULT; replaced final FC with 5 outputs
Initialization	Random, trained from scratch	ImageNet weights; frozen earlier layers
Optimization	Adam, lr=0.001, weight decay 0.0001	AdamW: head warm-up lr=0.001; layer4 lr=0.00003 and head lr=0.0003; wd=0.0001
Training schedule	Up to 15 epochs, early stopping patience 4	3 head-only epochs + up to 12 layer4/head epochs
Common setup	Batch size 64; CrossEntropyLoss; seed 42	Batch size 64; CrossEntropyLoss; seed 42
Input normalization	Mean/std=0.5 per RGB channel	ImageNet mean/std
Checkpoint criterion	Lowest validation loss	Lowest validation loss

The comparison is between two complete pipelines, not architecture depth alone: ResNet18 also differs in pretrained weights, normalization and optimizer. The notebook contains no training-time augmentation; the transformations below were applied for evaluation only.

4. Implementation

The executed Jupyter/PyTorch pipeline (1) downloaded/extracted the BOP archives; (2) parsed scene_gt.json and scene_gt_info.json, bounding boxes, camera poses, visibility and masks; (3) built reproducible scene-disjoint instance tables; (4) cropped, letterboxed and normalized RGB inputs; (5) trained both networks with validation-only checkpoint selection; (6) saved training histories, best .pt weights, prediction CSVs and confusion matrices; (7) evaluated deterministic stress conditions on validation images; (8) grouped errors by real BOP metadata; and (9) performed one final evaluation on the previously used test scenes and generated exportable outputs.

The validation robustness functions preserved deterministic perturbations across both models: in-plane rotations (15°/30°), image downscaling (75%/50%), gray patch occlusion (15%/30% of canvas area), and foreground-preserving insertion of 2/4 distractor patches taken from training images. Viewpoint change was additionally analyzed through 3D annotation directions; 2D image rotation was not treated as true 3D viewpoint change.

5. Results

Clean recognition performance and Week 07 comparison

Model / evidence	Validation accuracy	Test accuracy	Test macro-F1	Test errors
Historical Week 07 CNN	86.67%	72.67%	0.7132	123 / 450
Week 08 retrained CompactCNN	100.00%	88.00%	0.8826	54 / 450
Week 08 ResNet18	98.13%	75.78%	0.7772	109 / 450

The retrained CompactCNN selected its best checkpoint at epoch 5 (validation loss 0.00095); ResNet18 selected epoch 15 (validation loss 0.17414). On the repeated held-out scenes, the compact model was ahead of ResNet18 by 12.22 percentage points and 0.1054 macro-F1. The retrained compact score was 15.33 points above the historical Week 07 result; this is a different training run, not proof that a single architectural change caused the improvement.

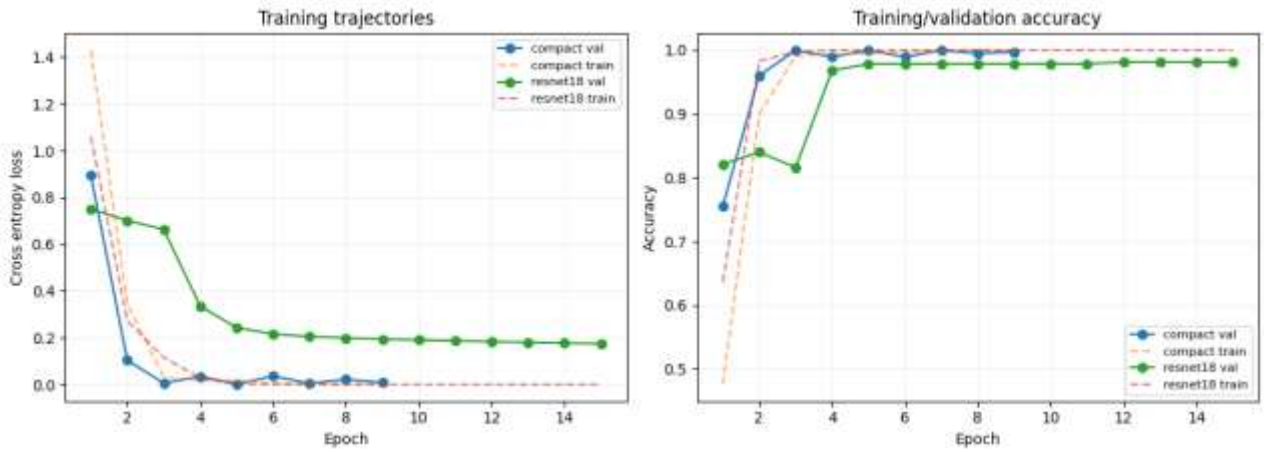


Figure — Recorded train/validation cross-entropy and accuracy for both Week 08 models.

Controlled robustness tests (375 validation samples per scenario)

Input condition	Compact accuracy	ResNet18 accuracy	Change vs clean: Compact / ResNet
Clean	100.00%	98.13%	0.00 / 0.00 pp
Rotate +15°	98.93%	62.93%	−1.07 / −35.20 pp
Rotate +30°	63.20%	52.80%	−36.80 / −45.33 pp
Scale to 75%	76.53%	21.87%	−23.47 / −76.27 pp
Scale to 50%	40.00%	20.00%	−60.00 / −78.13 pp
Occlude 15%	97.07%	77.33%	−2.93 / −20.80 pp
Occlude 30%	90.13%	51.47%	−9.87 / −46.67 pp
2 clutter patches	97.33%	87.47%	−2.67 / −10.67 pp
4 clutter patches	89.87%	82.40%	−10.13 / −15.73 pp

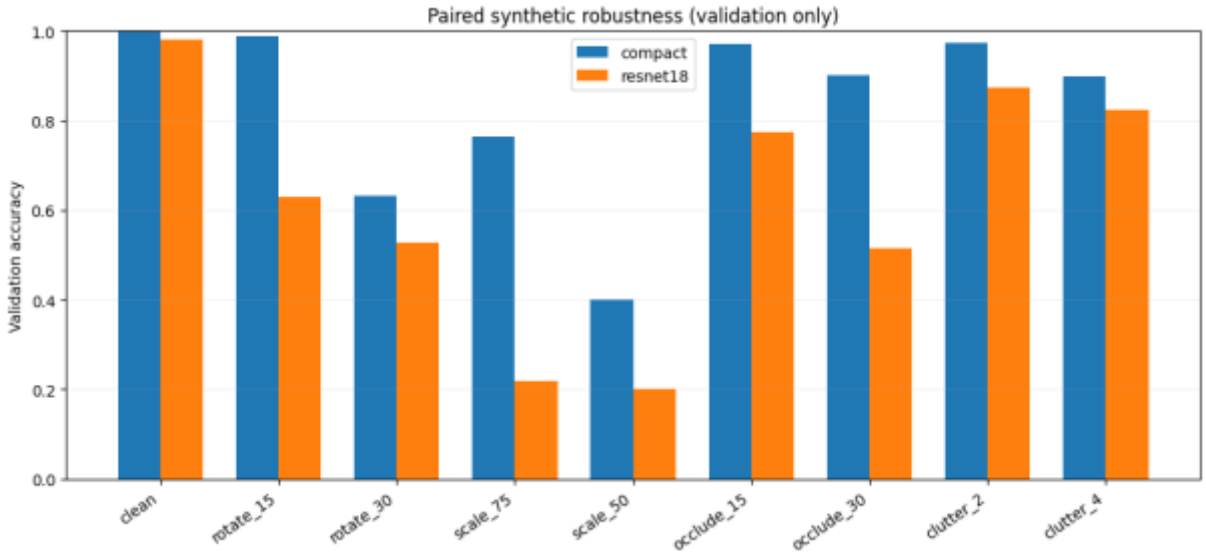


Figure — Paired validation accuracy under synthetic viewpoint proxies, scaling, occlusion and clutter.

ResNet18 showed the largest deterioration under scaling: accuracy fell to 21.87% at 75% scale and to 20.00% at 50% scale (chance level in a balanced five-class sample). CompactCNN also degraded under strong scaling and rotation, but less severely. These results measure synthetic stress sensitivity, not validated real-world performance.

Natural BOP metadata and representative outputs

Using camera-to-object pose, I measured each validation instance's minimum angle to a same-class training viewpoint. For ResNet18, accuracy was 99.13% (n=229) for 0–15°, 95.92% (n=98) for 15–30°, and 97.92% (n=48) for 30–60°. For natural image scale, it was 97.21% (n=215, medium) versus 99.38% (n=160, small). On visibility bins it was 98.18% (n=110, 0.50–0.75) and 98.11% (n=265, 0.75–1.00). All these subsets had 100% accuracy under CompactCNN.

The natural validation sample contains no high-angle (>60°) bin, no large-scale bin and no very-low-visibility (≤0.50) bin; its observed clutter counts also cover only 1 or 2+ nearby annotated objects. Therefore these strata do not establish comprehensive real-world robustness. Macro-F1 for subgroups missing some classes was excluded from interpretation because the notebook's fixed five-class macro averaging can produce deceptively low scores even at 100% subgroup accuracy.

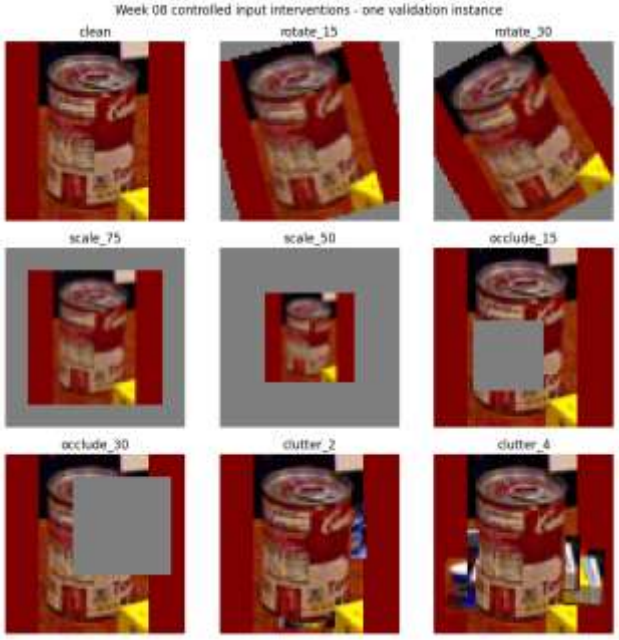


Figure — Actual validation-object input under all nine deterministic perturbation scenarios.

6. Failure Analysis

Validation errors: the retrained CompactCNN made 0/375 clean-validation mistakes, while ResNet18 made 7/375: master_chef_can → bleach_cleanser (4), bleach_cleanser → sugar_box (2), and tomato_soup_can → sugar_box (1). The error gallery depicts these three kinds of mistakes, including representative scene 55 crops. The examples show difficult partial views, highlights and background context; those visual factors are plausible contributors, not verified causal explanations.



Figure — Real ResNet18 validation mistakes (scene 55), extracted from the executed notebook.

Repeated test-scene failure patterns

Final repeated-test confusions were more substantial. CompactCNN classified 70/75 master_chef_can correctly (93.33% recall) but misclassified 24/75 sugar_box as tomato_soup_can. ResNet18 correctly classified only 40/75 master_chef_can (53.33% recall), assigning 35 to tomato_soup_can. It also sent 36/150 tomato_soup_can to master_chef_can and another 25 to tuna_fish_can. All 75 tuna_fish_can and all 75 bleach_cleanser test objects were correctly classified by ResNet18, while tomato_soup_can recall was only 80/150 (53.33%).

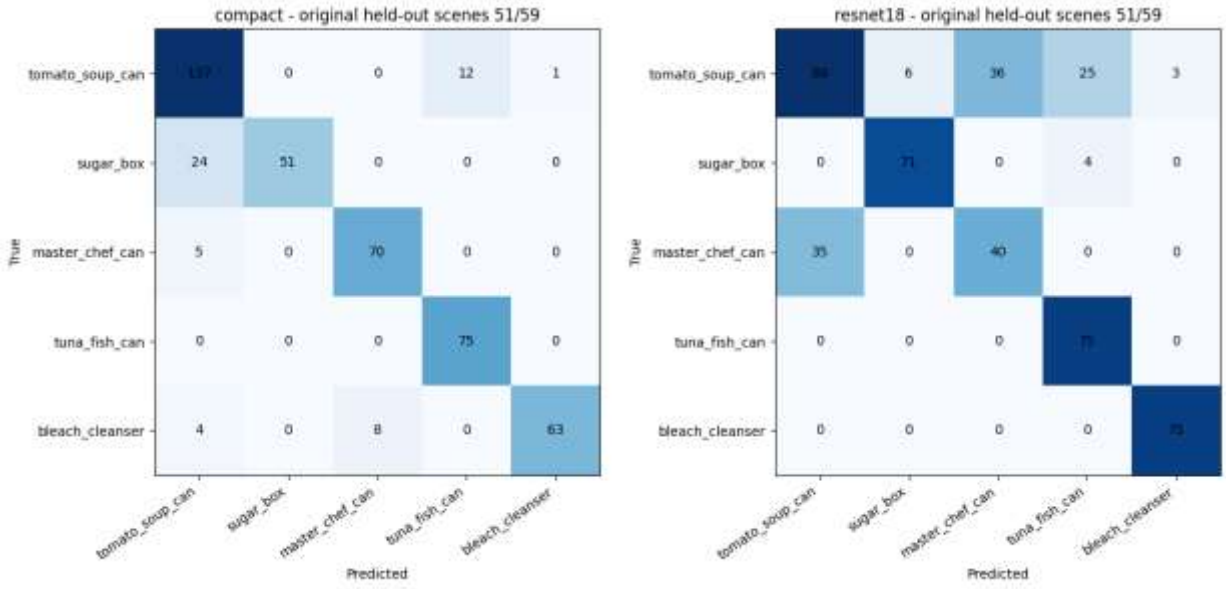


Figure — Measured test confusion matrices: retrained CompactCNN (left), ResNet18 (right).

My interpretation: the test errors are consistent with strong viewpoint/background dependence and confusion between visually related cylindrical packaging at a coarse 64×64 resolution. The large validation-to-test gaps (12.00 pp for CompactCNN, 22.36 pp for ResNet18) reinforce that near-perfect validation accuracy did not guarantee scene transfer. The synthetic interventions also exposed a potential ResNet18 sensitivity to input scale and occluding patches. These are evidence-based hypotheses, not a diagnosis of a single confirmed cause. Neither model was retrained in response to test errors.

7. Conclusion

This week demonstrated that a larger pretrained CNN is not automatically the best choice on a small, scene-dependent object subset. Despite a high clean validation score, ResNet18 performed worse than the retrained CompactCNN on the repeated test scenes and lost more accuracy under controlled scale, rotation and occlusion changes. The useful practices to keep are scene-disjoint evaluation, aspect-preserving crops, reproducible fixed splits, validation-only checkpoint selection, model-specific normalization and explicit class-wise error analysis. A simple network can remain a strong reference point.

For the next experiment, I would prioritize a fresh genuinely unseen scene split or external test; then test train-only multi-scale/rotation/occlusion augmentation, higher-resolution crops and controlled fine-tuning of earlier ResNet blocks. These should be compared one change at a time with multiple random seeds and focus on master_chef_can and tomato_soup_can recall. The assigned Week 09 task is to progress from ground-truth object crops to full-frame object detection, predicting class labels and bounding boxes and evaluating localization with Intersection over Union (IoU).